

Assignment 1

C programming Lab (Section C)

Instruction:

1. Follow the programming guidelines taught in the theory class. Refer to the provided slides to ensure you're writing in a clear and effective style.
 2. Once completed, show your code to the respective TAs.
 3. You may ask the TAs for help if you're encountering syntax errors or having trouble understanding the questions. However, the TAs will not provide you with the exact logic; you'll need to figure that out on your own.
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Q1. **Problem:**

Write a C program that simulates a **simple calculator**. The program should be able to perform basic arithmetic operations: addition, subtraction, multiplication, and division.

Input: The program should take two numbers as input from the user.

Output: The program should display the result of the operation.

Sample Example:

Enter first number: 10
Enter second number: 5
Sum is: 15
Subtraction is: 5
Multiplication is: 50
Division is: 2

File Name: assign01a.c

Q2. **Problem:**

Three friends are designing different parts of a garden:

1. **Cindy** plants flowers in a *circular* section of radius R units.
 - **Cost of soil** = ₹40 per square unit

- **Soil cost** = $\pi \times R^2 \times 40$
- 2. **Rick** creates a *rectangular* vegetable patch of length L units and width B units.
 - **Cost of fencing** = ₹16 per unit of boundary length
 - **Fence cost** = $2 \times (L + B) \times 16$
- 3. **Sam** builds a *square* sandbox with side S units.
 - **Cost of sand** = ₹24 per square unit
 - **Sand cost** = $S^2 \times 24$

File Name: assign01b.c

1. Test Case 1

- $R = 1, L = 2, B = 2, S = 1$
- Cindy's cost = $\pi \times 1^2 \times 40 \approx 125.66$ (using $\pi \approx 3.14159$)
- Rick's cost = $2 \times (2 + 2) \times 16 = 2 \times 4 \times 16 = 128$
- Sam's cost = $1^2 \times 24 = 24$

2. Test Case 2

- $R = 3, L = 4, B = 6, S = 5$
- Cindy's cost = $\pi \times 3^2 \times 40 \approx 3.14159 \times 9 \times 40 \approx 1130.97$
- Rick's cost = $2 \times (4 + 6) \times 16 = 2 \times 10 \times 16 = 320$
- Sam's cost = $5^2 \times 24 = 25 \times 24 = 600$

3. Test Case 3

- $R = 10, L = 15, B = 5, S = 8$
- Cindy's cost = $\pi \times 10^2 \times 40 \approx 3.14159 \times 100 \times 40 \approx 12566.37$
- Rick's cost = $2 \times (15 + 5) \times 16 = 2 \times 20 \times 16 = 640$
- Sam's cost = $8^2 \times 24 = 64 \times 24 = 1536$